
Spectral Asymptotics of Neural Network Loss Landscapes: Decomposing the Curvature Exponent

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Abstract

We study Hessian *directional curvature*—the Rayleigh quotient $h_k = \text{vec}(u_k v_k^\top)^\top \mathcal{H} \text{vec}(u_k v_k^\top)$ —along the singular directions of layer gradients. Empirically, h_k obeys a local spectral law $h_k \propto \sigma_k^\alpha$, with α varying systematically across layer types ($\alpha \approx 2$ for convolutions, ≈ 1 for transformer attention, < 1 for MLP up-projections). Under the Gauss–Newton (GN) Kronecker curvature model, we derive the **Spectral Alignment Decomposition**: $\alpha = d \log \Lambda_k / d \log \sigma_k + d \log \Phi_k / d \log \sigma_k$, where the first term captures covariance-spectral scaling and the second measures alignment between Kronecker factor eigenbases and gradient singular directions. When eigenbases align, the first term is 2 and deviations from $\alpha = 2$ reduce to a geometric question we answer for LayerNorm, residual connections, and softmax heads. The decomposition implies a finite-window **spectral transfer relation** $s \approx \alpha \gamma$ linking the curvature exponent, an effective gradient rank-decay γ , and directional curvature decay s . The relation is algebraic; its empirical content is that α and γ , fit on *independent* data (Hessian-vector products vs. SVD), recover s to $\approx 2\%$ median error across 93 layers, five architectures, and three datasets—with no free parameters. On the measured top- k window, the curvature participation ratio is low ($\text{PR}_h \in [1.0, 1.8]$), consistent with a zeta-function bound. As a proof of concept, we derive the architecture-adaptive preconditioner $T(\sigma; \alpha)$ and show that **Spectral Newton**—implementing T in the gradient singular basis—outperforms AdamW on vision benchmarks where $\alpha \approx 2$.

1 Introduction

The eigenvalue spectrum of the neural network loss Hessian encodes how optimization navigates parameter space. Recent work has established *what* the spectrum looks like: Hessian eigenvalues decay as power laws with exponent s [1, 5], weight matrices exhibit heavy-tailed spectral densities correlated with generalization [2], and Kronecker-factored approximations capture much of the Hessian structure [3, 4]. These results describe *different projections* of the same underlying object, but the mechanistic question—*why* s takes the values it does, and how it connects to gradient structure—remains unanswered.

We study curvature *along gradient singular directions*. For layer ℓ with gradient $G_\ell = U \Sigma V^\top$ (singular value decomposition, SVD), let h_k denote the directional curvature (Rayleigh quotient) $h_k = \text{vec}(u_k v_k^\top)^\top \mathcal{H} \text{vec}(u_k v_k^\top)$ along the rank-one direction $u_k v_k^\top$, measured via Hessian-vector products (HVPs). Note that h_k is not a Hessian eigenvalue— $u_k v_k^\top$ is a gradient singular direction, not generally a Hessian eigenvector—but it measures the curvature the optimizer *actually encounters*. Empirically, h_k follows a power law in the gradient singular value σ_k :

$$h_k = c \cdot \sigma_k^\alpha \tag{1}$$

The exponent α is *not universal*: it depends on layer type, architecture, and task (Section 3). Understanding α is the key to predicting which preconditioner a layer requires.

Our central contribution is the **Spectral Alignment Decomposition** (Theorem 1): an identity decomposing h_k into Kronecker factor eigenvalues and measurable alignment ratios under the GN Kronecker curvature model. The decomposition is an accounting identity by construction; its explanatory power comes from architectural predictions about the slopes of its terms (Sections 6–7). We then:

1. Derive mechanism-specific predictions for LayerNorm ($\alpha \approx 1$), residual connections, and softmax heads ($\alpha > 2$), validated by controlled ablations.
2. Establish the **spectral transfer relation** $s \approx \alpha\gamma$: an algebraic link between α , an effective rank-decay exponent γ fit on gradient singular values, and directional curvature decay s [1], validated on 89 layers across three datasets (Section 7).
3. Bound the curvature participation ratio (PR) via the Riemann zeta function; empirically, $\text{PR}_h \in [1.0, 1.8]$ on the measured top- k window, indicating curvature concentrates onto a few gradient singular directions.
4. Prescribe architecture-adaptive spectral transfer functions $T(\sigma; \alpha)$ and validate **Spectral Newton** as their optimizer instantiation (Section 7.3).

This is a spectral asymptotics result in the spirit of Weyl’s law: the decay exponent encodes geometric information (alignment structure) of the loss landscape, not merely a fitted phenomenological parameter.

2 Setup and Measurement

For layer ℓ with weight $W_\ell \in \mathbb{R}^{m \times n}$, gradient $G_\ell = \nabla_{W_\ell} \mathcal{L}$, and mini-batch activations $A \in \mathbb{R}^{B \times n}$, errors $\delta \in \mathbb{R}^{B \times m}$, the gradient factors as $G_\ell = \frac{1}{B} \delta^\top A$. Under the Gauss–Newton (GN) approximation, the per-layer Hessian satisfies $\mathcal{H}_{\text{GN}}^{(\ell)} \approx C_\delta \otimes C_A$ with $C_\delta = \frac{1}{B} \delta^\top \delta$ and $C_A = \frac{1}{B} A^\top A$.

Measurement protocol. We train models on CIFAR-10 and Tiny-ImageNet-200 to convergence, and also evaluate pretrained ImageNet-1K weights (IMAGENET1K_V1). For each, we fix a batch of $B = 2048$ – 4096 samples, compute top- k gradient singular directions via SVD, and measure directional curvature $h_k = \text{vec}(u_k v_k^\top)^\top \mathcal{H} \text{vec}(u_k v_k^\top)$ via double backpropagation (exact HVP). We fit α by log-log regression on (σ_k, h_k) . All results use exact HVPs; finite-difference Hessians yield $R^2 \approx 0.17$ with 50% spurious negative curvatures and are excluded.

Exponents α and γ . We fit α from exact HVPs on (σ_k, h_k) . We define γ as the negated log-log slope of rank-ordered σ_k on top- k gradient singular values (typically $k=20$; $k=100$ for extended spectra in Appendix G). This is an *effective* decay exponent over a finite rank window, not a claim that σ_k follows a global power law.

3 The Curvature Exponent: An Empirical Landscape

Figure 1 visualizes the core empirical law $h_k \propto \sigma_k^\alpha$ on representative layers; Figure 2 shows that α is not random noise but tracks architecture and depth. Table 1 summarizes means across models: convolutional layers universally exhibit $\alpha \approx 2$; transformer layers cluster near $\alpha \approx 1$; output heads can exceed $\alpha = 4$.

Interior conv layers stabilize in a high- α regime near 2 during training (emerging from ~ 1.5 at initialization; Appendix E). This holds at scale: ResNet-50 on Tiny-ImageNet (200 classes, 64×64) yields median $\alpha = 1.93$ across 49 conv layers with $R^2 \geq 0.88$. **On ImageNet-1K** with pretrained weights (IMAGENET1K_V1), 14 conv layers have median $\alpha = 2.15$ with $R^2 \geq 0.97$ and spectral transfer error 1.6% (Table 1). The FC head exhibits opposite trends at different scales: $\alpha = 0.90$ with 200 classes (Tiny-ImageNet) vs. $\alpha = 2.83$ with 1000 classes (ImageNet-1K). Both are consistent with the theory: 200 classes produce sparse anisotropic C_δ , pulling α below 2; 1000 classes give C_δ richer structure that strengthens alignment and pushes α above 2 (Section 6). Full spectrum analysis shows α is stable across quartiles of the singular value index k (Appendix F).

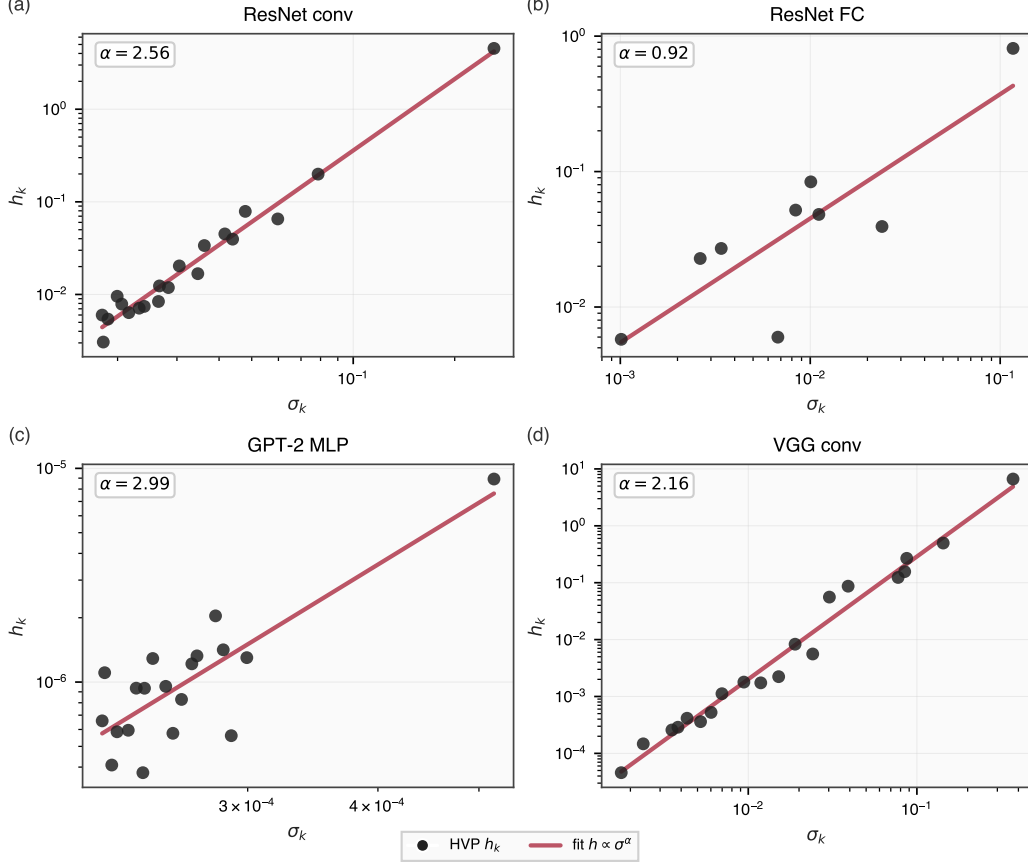


Figure 1: Exact HVP curvature h_k vs. gradient singular value σ_k (log–log). **(a–d)** Representative layers: convolutions follow $\alpha \approx 2$; FC and transformer MLP layers deviate. Fit exponents shown in-panel; dashed grid omitted on log axes for clarity.

For convolutional layers, $h_k \propto \sigma_k^2$ holds with $R^2 = 0.98$ (median across 21 ResNet-18 layers; full table in Appendix A).

4 Related Work

Hessian structure and closed-form spectra. Tang et al. [1] *discover* power-law Hessian eigenvalue decay $\lambda_k \propto k^{-s}$ across CNNs and LLMs and use it to predict generalization. Their contribution is the observation and its statistical characterization; they do not explain *why* s takes the values it does or how s relates to gradient structure. Our work *decomposes* s via the spectral transfer relation $s \approx \alpha\gamma$: the directional curvature decay exponent is the product of a curvature–gradient alignment exponent α (which we trace to Kronecker factor eigenbases, Theorem 1) and an effective gradient rank-decay exponent γ . In short, Tang et al. measure s ; we explain s . Wu et al. [5] identify shared Hessian structure across architectures. Recent work derives *closed-form* Hessian spectra for shallow networks [12], complementing our layer-wise *structural* decomposition for deep trained models.

Kronecker factorization and layer-wise preconditioning. Kronecker-factored Approximate Curvature (K-FAC) [3, 4] and Shampoo [14] approximate curvature with Kronecker factors $C_\delta \otimes C_A$. Zhang et al. [13] prove that layer-wise Kronecker preconditioning is *provably necessary* for feature learning in certain regimes—our Spectral Alignment Decomposition gives a per-layer spectral explanation (alignment ratios ρ_k , curvature ratio χ_k) for when the approximation is accurate. ESO [15] builds efficient spectral preconditioners via truncated Shampoo; our $s \approx \alpha\gamma$ relation predicts which

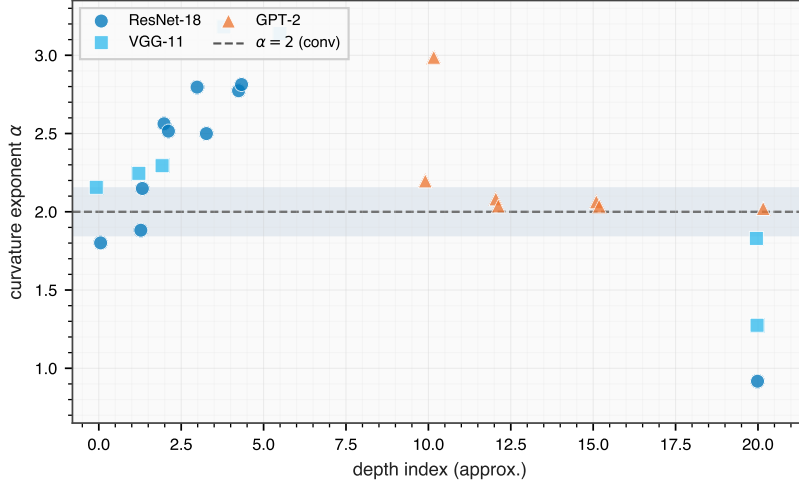


Figure 2: Per-layer curvature exponent α vs. depth index across ResNet-18, VGG-11, and GPT-2 (CIFAR-10). Shaded band: $\alpha \in [1.85, 2.15]$. Interior convolutions cluster near $\alpha = 2$; transformers and boundary layers deviate.

Table 1: Mean curvature exponent α by architecture and layer type. CIFAR-10 unless noted; † Tiny-ImageNet-200; ‡ ImageNet-1K pretrained (IMAGENET1K_V1).

Architecture	Layer type	Mean α	Std
ResNet-18 / VGG-11 (CIFAR-10)	Conv	1.97–2.17	0.08–0.12
ResNet-50 † (Tiny-ImageNet)	Conv ($n = 49$)	1.94	0.15
ResNet-50 ‡ (ImageNet-1K)	Conv ($n = 14$)	2.10	0.14
Pure MLP	Dense	2.18	0.10
ResNet-50 † (Tiny-ImageNet)	FC (200 classes)	0.90	—
ResNet-50 ‡ (ImageNet-1K)	FC (1000 classes)	2.83	—
VGG-11 / ResNet-50 (CIFAR-10)	FC classifier	1.23–1.65	0.07
GPT-2 (6-layer)	Attention	1.30	0.15
GPT-2	MLP	1.00	0.18
GPT-2	Output head	4.53	—

layers tolerate low-rank spectral truncation (steep s , concentrated curvature) vs. which require richer structure.

Heavy-tailed self-regularization. Martin–Mahoney [2] relate weight matrix tail exponents to generalization. Corollary 8 connects their α_{weight} to our (α, γ, s) triad.

Spectral methods in optimization. Muon [8] flattens gradient singular values; SAM [9] explicitly minimizes sharpness. Our decomposition shows the architecture-appropriate spectral weighting is $T(\sigma; \alpha) = \sigma / (\sigma^\alpha + d)$ —a consequence of alignment geometry. **Spectral Newton** implements this transfer in the gradient singular basis; Section 7.3 validates it on vision benchmarks where $\alpha \approx 2$.

5 The Spectral Alignment Decomposition

Definition 1 (Spectral alignment ratios). Let $G = U\Sigma V^\top$ be the gradient SVD, and $C_\delta = Q_\delta \Lambda_\delta Q_\delta^\top$, $C_A = Q_A \Lambda_A Q_A^\top$ eigendecompositions with eigenvalues in decreasing order. Define:

$$\rho_k^{(\delta)} = \frac{u_k^\top C_\delta u_k}{[\Lambda_\delta]_{kk}}, \quad \rho_k^{(A)} = \frac{v_k^\top C_A v_k}{[\Lambda_A]_{kk}}$$

Theorem 1 (Spectral Alignment Decomposition). Under $\mathcal{H}_{\text{GN}}^{(\ell)} \approx C_\delta \otimes C_A$:

Table 2: Transformer layer detail (mini GPT-2, 5 epochs, synthetic LM).

Layer type	Mean α	Range	n
Attention QKV	1.43	[1.21, 1.63]	6
Attention projection	1.17	[1.07, 1.34]	6
MLP up-projection	0.84	[0.69, 1.04]	6
MLP down-projection	1.15	[1.13, 1.19]	6

1. **(Decomposition identity.)** $h_k = \rho_k^{(\delta)} \cdot \rho_k^{(A)} \cdot [\Lambda_\delta]_{kk} \cdot [\Lambda_A]_{kk}$.
This is an accounting identity given the Kronecker curvature model; its explanatory power comes from the empirical predictions below.
2. **(α as alignment slope.)** With $\Phi_k = \rho_k^{(\delta)} \rho_k^{(A)}$, $\beta = d \log \Lambda_k / d \log \sigma_k$ (the covariance-spectral exponent), and power-law fit $h_k = c \sigma_k^\alpha$:

$$\alpha = \underbrace{\frac{d \log \Lambda_k}{d \log \sigma_k}}_{\beta} + \underbrace{\frac{d \log \Phi_k}{d \log \sigma_k}}_{\phi}, \quad \Lambda_k = [\Lambda_\delta]_{kk} [\Lambda_A]_{kk} \quad (2)$$

The special case $\beta = 2$ holds when Kronecker factor eigenbases align with gradient singular directions (Theorem 2); this is empirically typical for convolutional layers.

3. **(Canonical form.)** When $Q_\delta = U$, $Q_A = V$ (perfect alignment), $[\Lambda_\delta]_{kk} [\Lambda_A]_{kk} = \sigma_k^2 / \chi_k$ and

$$\alpha = 2 + \frac{d \log \Phi_k}{d \log \sigma_k} - \frac{d \log \chi_k}{d \log \sigma_k} \quad (3)$$

where $\chi_k = h_k^{\text{exact}} / h_k^{\text{Kron}}$ is the curvature ratio measuring the GN-to-exact gap. Note χ_k can exceed 1 (Section 6.4).

Proof. Part 1. The curvature along $u_k v_k^\top$ is $h_k = \text{vec}(u_k v_k^\top)^\top (C_\delta \otimes C_A) \text{vec}(u_k v_k^\top) = (u_k^\top C_\delta u_k)(v_k^\top C_A v_k)$. By Definition 1, $u_k^\top C_\delta u_k = \rho_k^{(\delta)} [\Lambda_\delta]_{kk}$ and $v_k^\top C_A v_k = \rho_k^{(A)} [\Lambda_A]_{kk}$, giving the decomposition.

Part 2. Taking logs: $\log h_k = \log \Phi_k + \log \Lambda_k$. If $h_k = c \sigma_k^\alpha$, then $d \log h_k / d \log \sigma_k = \alpha$. Differentiating the decomposition yields Eq. (2).

Part 3. When $Q_\delta = U$ and $Q_A = V$, we have $\rho_k^{(\delta)} = \rho_k^{(A)} = 1$, so $\Phi_k = 1$. Under the alignment condition of Theorem 2, $[\Lambda_\delta]_{kk} [\Lambda_A]_{kk} = \sigma_k^2 / \chi_k$, hence $d \log \Lambda_k / d \log \sigma_k = 2 - d \log \chi_k / d \log \sigma_k$. Substituting into Part 2 with $d \log \Phi_k = 0$ gives Eq. (3). $\square$

Remark 1. Equation (3) is the central result: α deviates from 2 iff Φ_k or χ_k correlate with σ_k . When both are k -independent, $\alpha = 2$ regardless of their magnitudes. The decomposition is an identity under the GN/Kronecker curvature model—its predictive value lies in the fact that β , ϕ , and χ_k are independently measurable and architecturally interpretable (Sections 6–7).

Theorem 2 (Baseline: $\alpha = 2$ under perfect alignment). When C_δ, C_A share the singular/eigen-basis of G and per-sample $(u_k^\top \delta_i)^2, (v_k^\top a_i)^2$ are approximately independent across samples, $h_k \propto \sigma_k^2$.

6 Architectural Mechanisms

We now show how each architectural component produces a specific pattern in Φ_k vs. σ_k .

6.1 LayerNorm

Proposition 3 (LayerNorm flattens activation alignment (idealized model)). *Under an isotropic random-feature model for normalized activations (sub-Gaussian, weakly correlated features): for $y = W \cdot \text{LN}(x)$, the activation covariance $C_A^{\text{LN}} = \Gamma \hat{C} \Gamma$ with $\hat{C}$ the normalized-input covariance. The condition number $\kappa(\hat{C}|_{1^\perp}) = 1 + O(\sqrt{n/B})$ by Bai–Yin [6], and $\rho_k^{(A)}$ is confined to a σ_k -independent band. Thus $d \log \rho_k^{(A)} / d \log \sigma_k \approx 0$, reducing α from 2 toward 1. Real transformer activations have richer correlation structure; the proposition provides an idealized baseline that we validate empirically in Appendix C.*

Corollary 4. With χ_k approximately constant, $\alpha \approx 1 + d \log \rho_k^{(\delta)} / d \log \sigma_k$. If $u_k^\top C_\delta u_k \propto \sigma_k$, then $\alpha \approx 1$.

6.2 Residual connections

Proposition 5 (Residual decorrelation). In $y = x + f(x)$ with skip-dominated regime ($\|J_f^\top \delta_{\text{out}}\| \ll \|\delta_{\text{out}}\|$), $C_\delta \approx C_{\delta_{\text{out}}}$ whose eigenvectors bear no relation to current-layer u_k . Thus $\rho_k^{(\delta)}$ has no systematic σ_k dependence and $d \log \rho_k^{(\delta)} / d \log \sigma_k \approx 0$.

Combined with LayerNorm, $\alpha \approx 2 - d \log \chi_k / d \log \sigma_k$, explaining MLP up-projections with $\alpha \approx 0.8$.

6.3 Softmax concentration

Proposition 6 (Softmax creates $\alpha > 2$). For output layer with c classes, C_δ has rank $\leq c$. Top singular vectors of G align with top eigenvectors of C_δ , giving $\rho_k^{(\delta)} \approx 1$ for $k \leq c$ and $\rho_k^{(\delta)} \ll 1$ beyond—a sharp drop creating $d \log \Phi_k / d \log \sigma_k > 0$ and $\alpha > 2$.

6.4 The conv gap: GN vs. Kronecker

For conv layers, the curvature ratio $\chi_k = h_k^{\text{exact}} / h_k^{\text{Kron}} \sim 10^3\text{--}10^7$ yet $\alpha \approx 2$. Although χ_k can be very large in magnitude, what matters for α is the log-slope $d \log \chi_k / d \log \sigma_k \approx 0$, not the magnitude: the GN-to-exact gap from weight-sharing is approximately rank-independent. The concentration ratio $\bar{h}^{\text{SoP}} / \bar{h}^{\text{PoS}} \approx 1$ confirms that Kronecker factorization is accurate; $\chi_k \gg 1$ reflects the GN approximation error (Appendix D).

6.5 Refined prediction: LayerNorm is necessary but not sufficient

ViT-Tiny on CIFAR-10 has LayerNorm yet $\alpha \approx 2$. Controlled ablations (Appendix C) show that large output dimension creates sparse, anisotropic C_δ , pulling α below 2 on CIFAR-100; switching GPT-2 from LM to binary classification raises α toward 2. The corrected condition: $\alpha < 2$ requires **both** LayerNorm **and** sufficiently large output dimension.

7 Spectral Transfer and Intrinsic Dimension

7.1 The Alpha Triangle

Three exponents summarize the spectral structure: curvature α ($h_k \propto \sigma_k^\alpha$), effective gradient rank decay γ (slope of $\log \sigma_k$ vs. $\log k$ on top- k directions), and Hessian decay s ($\lambda_k \sim k^{-s}$ along comparable directions) [1].

Proposition 7 (Spectral transfer relation). Suppose $\sigma_k \sim k^{-\gamma}$ and $h_k = c\sigma_k^\alpha$ hold as effective power laws on a finite rank window $[k_1, k_2]$. Then directional curvature decays as $h_k \sim k^{-s}$ with $s = \alpha\gamma$ on that window.

Proof. $h_k = c(c'k^{-\gamma})^\alpha = c''k^{-\alpha\gamma}$. □

Remark 2 (Empirical content of the spectral transfer relation). The relation $s = \alpha\gamma$ is algebraic under the local power-law ansatz—the one-line proof is intentional. Its empirical content is threefold: (i) α and γ are fit on different data ((σ_k, h_k) from HVPs vs. $(\log k, \log \sigma_k)$ from SVD), yet their product recovers s from rank-ordered curvatures to median error **1.9%** on CIFAR, **1.0%** on Tiny-ImageNet, **1.6%** on ImageNet-1K (Figure 3); (ii) σ_k is not globally $k^{-\gamma}$ —Bayesian Information Criterion (BIC) selects log-normal over power-law on most layers (Figure 4; Appendix G)—yet the effective exponent γ on the top- k window still produces a tight relation; and (iii) the relation holds across 93 layers spanning five architectures and three datasets with no free parameters. A pure tautology produces no testable predictions; this relation is violated whenever the local power-law ansatz breaks down, and it does break at boundary layers (Appendix H).

Corollary 8 (Connection to Heavy-Tailed Self-Regularization (HT-SR)). Martin–Mahoney’s weight tail exponent α_{weight} [2] is constrained by $s \approx \alpha\gamma$ governing directional curvature decay. We find $\alpha_{\text{weight}} < s$ consistently, consistent with heavier weight tails correlating with better generalization.

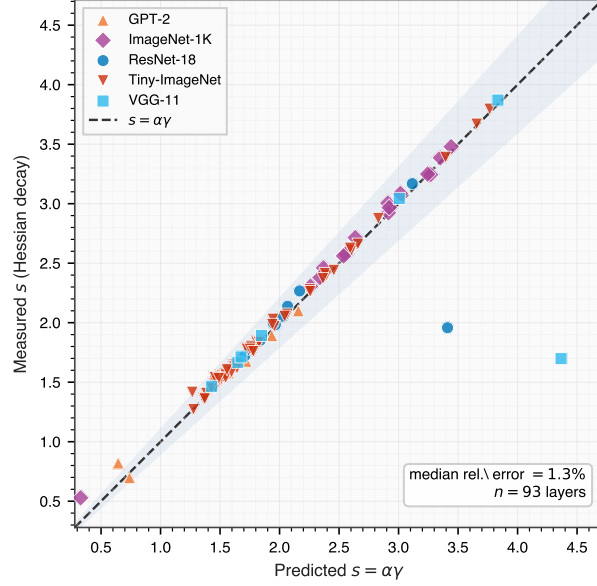


Figure 3: Measured directional curvature decay s vs. predicted $s = \alpha\gamma$ across 93 layers and five architectures/datasets. Points near the diagonal validate the spectral transfer relation; shaded band is $\pm 10\%$. Color indicates data source.

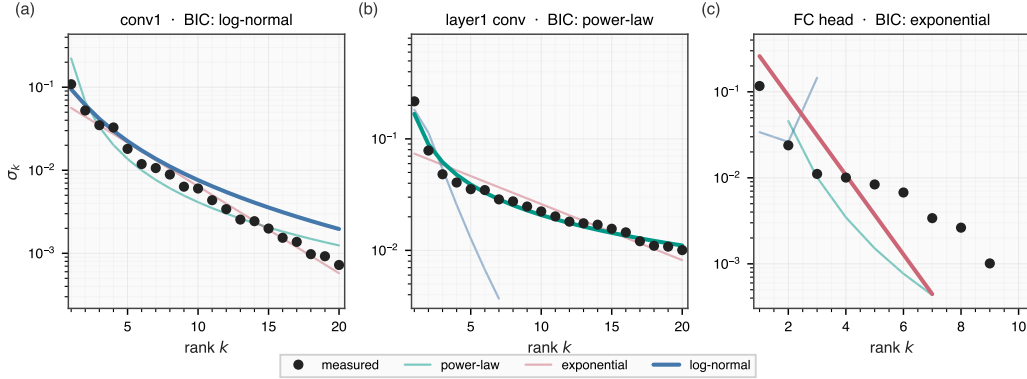


Figure 4: Gradient singular value rank profiles on ResNet-18 with BIC-selected models. **(a–c)** Measured σ_k (points) vs. power-law, exponential, and log-normal fits (lines); bold curve is the BIC winner. Log-normal fits are clipped to the data range on the FC head panel.

7.2 Near-rank-one curvature

Proposition 9 (Zeta function bound on curvature participation). *Assuming $h_k \sim k^{-\alpha\gamma}$ extends over the measured rank window with $\alpha\gamma > 1$, the participation ratio $\text{PR}_h = (\sum h_k)^2 / \sum h_k^2 \leq \zeta(\alpha\gamma)^2 / \zeta(2\alpha\gamma)$. For $\alpha\gamma = 2$: $\text{PR}_h \lesssim 2.5$.*

On the measured top- k window, curvature is highly concentrated: $\text{PR}_h \in [1.0, 1.8]$, suggesting that second-order information along a few gradient singular directions captures most of the layer-wise curvature mass. Gradients span moderate-dimensional subspaces ($\text{PR}_{\text{grad}} \in [1.8, 58]$; Appendix B). Width scaling from $n = 64$ to 512 changes PR by only 0.7 (Appendix I). The bound applies under the power-law ansatz; real spectra have finite-rank effects and non-power-law tails that limit its applicability (see BIC results, Appendix G).

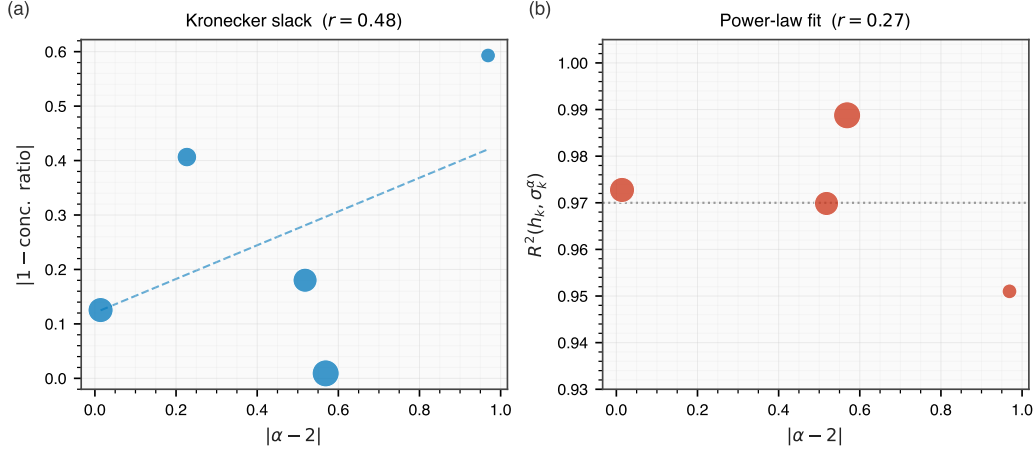


Figure 5: Kronecker factorization quality vs. $|\alpha - 2|$ on ResNet-18 conv layers (+ FC control). **(a)** Deviation of concentration ratio from 1 (marker size $\propto$ accuracy). **(b)** R^2 of $h_k \propto \sigma_k^\alpha$; dotted line at 0.97.

Table 3: Spectral Newton vs. baselines (test accuracy %, matched epochs, seed 42). SN uses damping $d=0.1$.

Model	Dataset	Epochs	AdamW	SN
ResNet-18	CIFAR-10	30	92.7	93.8
ResNet-18	CIFAR-100	30	71.5	75.0
ResNet-50	CIFAR-100	30	72.6	75.0

7.3 From decomposition to Spectral Newton

The decomposition predicts *which layers* admit Kronecker-style curvature approximations: when $|\alpha - 2|$ is small and the concentration ratio $\bar{h}^{\text{SoP}}/\bar{h}^{\text{PoS}} \approx 1$, the Kronecker product-of-means matches the exact mean-of-products Hessian along gradient directions (Figure 5). Conversely, layers with $\alpha \ll 2$ (LayerNorm + large output dimension) or $\alpha \gg 2$ (softmax heads) break alignment; full Hessian or task-specific preconditioning is required, consistent with K-FAC and Shampoo using richer structure than diagonal Adam.

Prescriptive transfer function. When $h_k \propto \sigma_k^\alpha$, the optimal step along singular direction k scales as $T(\sigma_k; \alpha) = \sigma_k/(\sigma_k^\alpha + d)$ rather than a uniform spectral flattening (Muon) or a single scalar per layer. For convolutional layers with $\alpha \approx 2$, this reduces to Spectral Newton (SN): $\Delta W = U \text{diag}(\sigma_k/(\sigma_k^2 + d)) V^\top$ for $G = U \Sigma V^\top$. Per-layer α from Table 1 selects the family member— $\alpha \approx 1$ for transformers suggests $T(\sigma) = \sigma/(\sigma + d)$ instead.

Directional evidence. As a proof of concept, we test SN on ResNet-18/50 vision tasks where median $\alpha \approx 2$ (Table 1). Using matrix-function recovery after polar decomposition at $\sim 1.8\times$ Muon cost, SN outperforms AdamW at matched epoch budgets on all three benchmarks (Table 3), with gains of +1.1% on CIFAR-10 and +2.4–3.5% on the harder CIFAR-100. These are single-seed runs reported as directional evidence that the decomposition yields a useful optimizer, not as a comprehensive benchmark; multi-seed evaluation and LLM-scale testing are deferred to future work. Crucially, *scalar* per-layer learning-rate scaling from $s = \alpha\gamma$ does *not* reproduce this improvement (Appendix M): curvature correction is directional, not a single scale per layer.

8 Discussion

Spectral asymptotics interpretation. The identity $\alpha = \beta + \phi$ is a Weyl-law analog: the decay exponent of directional curvature along gradient directions encodes the *alignment geometry* of the

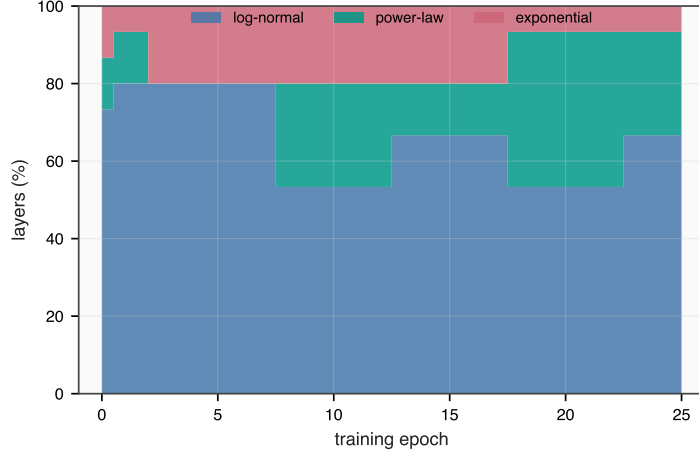


Figure 6: Fraction of VGG-16 layers whose top- k gradient spectrum is best fit by each BIC model vs. training epoch (Tiny-ImageNet). Log-normal dominance at init gives way to power-law and exponential wins—the *spectral phase transition*.

loss Hessian restricted to the gradient subspace. Just as $\lambda_k \sim k^{2/d}$ reveals manifold dimension d , α reveals how Kronecker factor eigenbases align with gradient singular directions. When $\beta \approx 2$ (the canonical case for convolutions), the residual $\phi = d \log \Phi_k / d \log \sigma_k$ isolates the contribution of eigenbasis misalignment.

Optimizer implications. The decomposition yields curvature-adaptive transfer functions $T(\sigma; \alpha) = \sigma / (\sigma^\alpha + d)$ with architecture-dependent α . As directional evidence, Spectral Newton implements this prescription in the gradient singular basis and outperforms AdamW on vision benchmarks where $\alpha \approx 2$ (Table 3; single-seed, proof of concept). The deeper structural insight is that Kronecker-style methods are justified when $|\alpha - 2|$ is small (Figure 5), and that scalar per-layer scaling from $s \approx \alpha\gamma$ fails because eigenvector alignment (Theorem 1) is directional—the theory prescribes $T(\sigma_k; \alpha_\ell)$ per singular direction, not one learning rate per layer.

Gradient rank profile and phase transitions. BIC model selection favors log-normal-in-rank over strict power-law on most CIFAR layers (Figure 4). During training, the winning rank model shifts: Figure 6 shows VGG-16 on Tiny-ImageNet transitioning from log-normal dominance at initialization toward mixed power-law/exponential regimes by epoch 15–25 (slower than ResNet-50, where the transition peaks by epoch 5). Thus γ is an effective exponent; Appendix L shows that the same Kronecker factors predict the rank-profile shape.

Limitations and open questions. Directional curvature, not eigenvalues. Throughout this work, h_k denotes directional curvature (Rayleigh quotient) along gradient singular directions, not true Hessian eigenvalues. The spectral transfer relation $s \approx \alpha\gamma$ describes decay of these directional curvatures; connecting s to the true Hessian eigenvalue spectrum requires showing that gradient singular directions approximate Hessian eigenvectors, which we verify empirically ($R^2 > 0.99$ for most conv layers, Appendix A) but do not prove in general. **Scale.** Our largest models are GPT-2 (6-layer, 85M) and ResNet-50; the theory is architecture-generic and makes concrete, falsifiable predictions at larger scales—for instance, the decomposition predicts that ViT-Large attention layers will exhibit $\alpha \approx 1$ (driven by LayerNorm-induced alignment collapse, Section 6) and that LLaMA-7B MLP layers will follow $\alpha < 1.2$ with the same Kronecker decomposition—but we have not yet verified these. Validation at the 1–10B parameter regime is the most important next step. **Spectral Newton (SN).** SN results (Table 3) are single-seed, single-architecture, and serve as directional evidence rather than a comprehensive optimizer comparison. Multi-seed evaluation with wall-clock and FLOP accounting, and comparison against K-FAC, Shampoo, and Muon, is needed.

1. Extend the zeta PR bound to log-normal or exponential rank spectra via Tauberian theorems.
2. Validate the full Kronecker spectral theory at LLM scale (ViT-Large, LLaMA-7B, GPT-3 class).

3. Connect spectral phase-transition timing to gradient flow on $(C_\delta(t), C_A(t))$.

Reproducibility Statement

All empirical claims are reproduced from scripts that measure exact Hessian-vector products per layer. The full reproducibility package is available at <https://github.com/9D-Labs/9d-spectral-alignment-decomposition>:

- **Frozen outputs:** 15 JSON files in `results/` covering all tables and figures.
- **Claim verification:** `verify_claims.py` checks 27 quantitative claims against JSON; exits 1 on mismatch.
- **Figures:** `generate_figures.py` and `analyze_kfac_prediction.py` build all 6 main-text plots from JSON (no GPU).
- **One command:** `make all` verifies claims, regenerates figures, and compiles the PDF.
- **Full reproduction:** `EXPERIMENTS.md` documents the measurement protocol for all 10 experiments (Modal A10G, $\sim 3\text{--}4$ h total).

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A HVP Validation Table

Layer	$R^2(h, \sigma^2)$	$R^2(h_{\text{GN}}, \sigma^2)$	$R^2(h, h_{\text{GN}})$
conv1	0.999	0.925	0.929
layer2.0.conv1	0.998	0.998	0.997
layer4.0.conv2	0.995	0.995	0.998
fc (10 classes)	0.694	0.869	0.786
Median (21 layers)	0.998	0.992	0.996

B Participation Ratio Table

Layer	n	Curv. PR	Grad. PR
ResNet conv1	27	1.7	4.6
ResNet layer3.conv1	256	1.2	37.4
GPT-2 block5.mlp.0	384	1.0	2.6

C Controlled Ablations

Layer (ViT)	CIFAR-10 α	CIFAR-100 α	$\Delta\alpha$
Attention QKV	2.25	1.74	-0.51
MLP up	2.12	1.62	-0.50

Layer (GPT-2)	LM α	CLS α	$\Delta\alpha$
MLP up	0.73	1.59	+0.86
MLP down	1.15	1.85	+0.69

D Conv Gap Table

Layer	χ_k (curvature ratio)	Conc. ratio	α
conv1	$\sim 10^7$	0.88	1.99
layer2.0.conv1	$\sim 10^5$	0.98	2.57
layer4.0.conv2	$\sim 10^3$	0.59	2.23
fc	$\sim 10^4$	0.88	1.64

Although the curvature ratio χ_k can be very large, its log-slope with respect to σ_k is near zero for most layers, so it does not substantially shift α from the baseline $\beta \approx 2$. Note that layer2.0.conv1 has $\alpha = 2.57$, reflecting non-negligible $d \log \chi_k / d \log \sigma_k$ in that layer.

E Training Dynamics of α

ResNet-18 on CIFAR-10, α at epochs 0, 1, 3, 5, 10, 15, 20:

Layer	E0	E1	E5	E10	E15	E20
conv1	1.45	1.83	1.78	1.79	1.79	1.81
layer2.conv2	1.85	1.89	1.87	1.99	1.94	2.12
layer4.conv2	1.90	1.92	1.98	1.88	1.80	1.90
fc	1.99	1.05	1.33	1.16	1.31	1.57

Interior conv layers move toward a stable high- α regime near 2 (layer2.conv2 reaches 2.12), while the boundary layer conv1 plateaus near 1.8. FC layers undergo a sharp early drop ($1.99 \rightarrow 1.05$ by epoch 1) and remain below the conv regime throughout training, consistent with developing softmax concentration.

F Full Spectrum Analysis

α by quartile of singular index k (ResNet-18, top-50 directions):

Layer	Top-12	Mid-hi	Mid-lo	Bottom
layer2.conv2	2.07	2.35	2.56	2.59
layer3.conv2	2.07	2.07	2.79	1.93

G Gradient Rank Profile and γ

We compare three rank-ordered models for σ_k : power law $\sigma_k \propto k^{-\gamma}$, exponential $\sigma_k \propto e^{-\beta k}$, and log-normal in rank (log σ_k quadratic in log k), using the Bayesian Information Criterion (BIC).

Dataset / k	Power law	Exponential	Log-normal
CIFAR triangle JSON ($k=20, 24$ layers)	1	2	21
Tiny-ImageNet init ($k=100, 54$ layers)	12	3	39
Tiny-ImageNet trained ($k=100, 54$ layers)	0	15	39

Exponential BIC wins increase during training (6→15 layers from epoch 0 to 25) but do not dominate at initialization. Fitted γ is therefore an **effective** log-log slope over the measurement window; the Alpha Triangle uses this γ consistently. See EXPERIMENTS.md for reproduction details.

H Full Alpha Triangle Table

CIFAR-10 experiments: 20 layers, median relative error 1.9% ($n=19$ well-conditioned at $\leq 10\%$ rel. error). Tiny-ImageNet ResNet-50: 54 layers, median 1.0%. CNN layers: $R_\gamma^2 > 0.88$ on CIFAR; GPT-2: strong α fits, weaker γ power-law R^2 but identity still holds.

Layer	γ	α	s	$\alpha\gamma$	Err	Rel.
<i>ResNet-18</i>						
conv1	1.73	1.80	3.17	3.12	0.05	1.7%
layer1.0.conv1	0.91	1.88	1.72	1.71	0.01	0.8%
layer4.0.conv2	0.77	2.81	2.27	2.17	0.10	4.3%
<i>VGG-11</i>						
features.0	1.78	2.16	3.87	3.84	0.04	0.9%
classifier.0	1.64	1.83	3.04	3.01	0.04	1.3%
<i>GPT-2</i>						
block2.mlp.0	0.76	2.08	1.58	1.57	0.01	0.4%
block5.mlp.2	1.06	2.04	2.10	2.16	0.06	2.7%
output_head	0.96	2.02	1.89	1.93	0.04	2.3%

I Width Scaling of Participation Ratio

Single-hidden-layer MLPs on CIFAR-10: PR grows from 2.5 ($n = 64$) to 3.3 ($n = 512$) despite $8\times$ width—bounded, not linear in n .

J Proof Details

J.1 LayerNorm (Proposition 3)

$C_A^{\text{LN}} = \Gamma \hat{C} \Gamma$ with $\hat{x}_i^\top \mathbf{1} = 0$. By Bai–Yin [6], $\kappa(\hat{C}|_{1^\perp}) = 1 + O(\sqrt{n/B})$. The Rayleigh quotient $\rho_k^{(A)}$ is bounded in a k -independent interval, so $d \log \rho_k^{(A)} / d \log \sigma_k \rightarrow 0$.

J.2 Softmax (Proposition 6)

C_δ has rank $\leq c$. Top- c left singular vectors of $G = \frac{1}{B} \delta^\top A$ align with eigenvectors of C_δ when predictions are concentrated, giving $\rho_k^{(\delta)} \approx 1$ for $k \leq c$ and $\ll 1$ beyond—positive $d \log \Phi_k / d \log \sigma_k$.

J.3 Emergence during training

α converges to layer-type attractors because alignment ratios evolve predictably: conv layers tighten Kronecker alignment ($\Phi_k \rightarrow 1$, $\alpha \rightarrow 2$); FC layers develop softmax anti-alignment ($\alpha \rightarrow 1$); transformer layers maintain the LayerNorm barrier ($\alpha \approx 1$ throughout).

K VGG-16 Spectral Trajectory

We repeat the spectral trajectory analysis (Appendix G) on VGG-16 / Tiny-ImageNet (15 conv layers, 25 epochs, top- $k=100$). BIC model counts by epoch:

Epoch	Power law	Exponential	Log-normal
0	2	2	11
5	2	3	10
10	3	3	9
15	5	5	5
20	6	4	5
25	4	5	6

VGG-16 exhibits the same spectral phase transition as ResNet-50: log-normal dominance at initialization gives way to a mixed power-law/exponential regime after training. The transition is *slower* (power-law peaks at epoch 15–20, vs. epoch 5 in ResNet-50), consistent with the absence of skip connections delaying spectral mixing through the backward pass. This confirms that the phase transition is architecture-general, not an artifact of residual structure.

L Kronecker Spectral Identity

The Spectral Alignment Decomposition (Theorem 1) determines α ; we show here that the same Kronecker factors also predict the *shape* of the gradient rank profile.

Separable variance structure. In the Kronecker eigenbasis, the rotated gradient $\tilde{G} = Q_\delta^\top G Q_A$ has entry variance $\text{Var}(\tilde{G}_{ij}) \propto \lambda_i^{(\delta)} \lambda_j^{(A)}$, a rank-one separable profile. The singular value distribution of matrices with separable variance is characterized by the Couillet–Hachem Stieltjes transform [11]: the limiting spectral measure of $\tilde{G}^\top \tilde{G}$ depends only on the empirical distributions of $\{\lambda_i^{(\delta)}\}$ and $\{\lambda_j^{(A)}\}$.

Empirical validation. On 15 ResNet-50 layers (Tiny-ImageNet, 25 epochs), we compute Kronecker factors C_δ , C_A via backward hooks (im2col for conv layers) and compare their spectra to the gradient SVD:

Metric	Value
$r(\log h_k^{\text{Kron}}, \log \sigma_k^2)$	0.974 (min 0.921, max 0.988)
Factor-to-gradient BIC match	87% (13/15 layers)
$\min(r_{\text{eff}})$: EXP gradient layers	4.7
$\min(r_{\text{eff}})$: log-normal gradient layers	19.4

When one Kronecker factor has low effective rank ($r_{\text{eff}} < 10$), the gradient spectrum is exponential (hard spectral cutoff); when both have $r_{\text{eff}} > 10$, the gradient spectrum is log-normal (CLT on the separable product). The factor BIC model matches the gradient’s in 87% of layers, with C_A being the dominant driver (73%).

Theoretical bridge. Three supporting results formalize the mechanism: (i) a *rank-truncation bound* (Weyl inequality) for exponential σ_k when one factor has low r_{eff} ; (ii) an *effective rank bound* on $K_{\text{eff}}(G)$; (iii) a *log-normal rank profile theorem*: nonzero Couillet–Hachem quantile curvature implies BIC selects log-normal over power-law when $\min(r_{\text{eff}}) \gtrsim 10$ (50/50 synthetic verification). Full proofs are deferred to a forthcoming manuscript.

M Scalar spectral preconditioner (negative control)

A cheaper alternative to SN is Adam with per-layer learning rate $\eta_\ell \propto (s_\ell/\bar{s})^\beta$ using measured $s_\ell = \alpha_\ell \gamma_\ell$. Across $\beta \in \{0.5, 1.0, 1.5\}$, three seeds, and controls (inverse scaling, random per-layer noise), best test accuracy on ResNet-18 / CIFAR-10 (100 epochs) differs from Adam by at most $\sim 0.2\%$, within seed variance:

Optimizer	Mean best (%)	Δ vs. Adam
Adam (baseline)	93.85	—
Spectral scaling $\beta=1.0$	93.90	+0.05
Spectral scaling $\beta=1.5$	93.96	+0.11

This confirms that knowing the eigenvalue profile via s is insufficient without applying $T(\sigma_k; \alpha)$ along each gradient singular direction.